MAINTENANCE TASK PRIORITY RANKING ALGORITHM

University Maintenance Monitoring and Management Platform

Priority Score Formula

PS = CW + SW + 3S + 3U + 4SR + 2AP + 2LI + WT

Symbol	Factor	Range
PS	Priority Score	Calculated
CW	Category Weight	3 - 10
SW	Subcategory Weight	1 - 5
S	Severity	1 - 5
U	Urgency	1 - 5
SR	Safety Risk	0 - 5
AP	Affected People	1 - 5
LI	Location Importance	1.5 - 10
WT	Waiting Time	1 - 5

Category Weight

Main category	CW
Electricity	10
Water	9
Civil & Structural	8
HVAC	7
Academic	5
Cleaning	4
Gardening	3

Subcategory Weight

Category	Subcategory	SW
Electricity	Electrical Panels; Power Supply; Wiring & Cables	5
Electricity	Switches & Sockets	4
Electricity	Ceiling Fans	3
Electricity	Lighting & Fixtures	2
Water	Water Pumps; Water Supply	5
Water	Pipelines & Leaks; Toilets & Flushing; Drainage & Overflow	4
Water	Taps & Valves	3
Civil & Structural	Structural Damage; Ceilings & Roofs; Elevator	5
Civil & Structural	Walls & Plaster; Flooring & Tiles; Doors & Windows	3
Civil & Structural	Furniture & Desks	2
HVAC	Chiller Systems	5
HVAC	Air Conditioning Unit; Duct Ventilation	4
HVAC	Exhaust Fans	3
HVAC	Thermostat Control; Filter Replacement	2
Academic	Laboratory Equipment	5
Academic	Classroom Equipment	4
Academic	Projectors & Displays	3
Academic	Whiteboards; Academic Furniture	2
Cleaning	Hazardous Spill	5
Cleaning	Toilet & Sanitation; Waste Management	4
Cleaning	Classroom Cleaning	3
Cleaning	Corridor Cleaning	2
Cleaning	General Cleaning	1
Gardening	Fallen/Dangerous Tree; Dangerous Branches	5
Gardening	Drainage/Vegetation Obstruction	4
Gardening	Tree Maintenance	3
Gardening	Grass Cutting	2
Gardening	General Landscaping	1

Severity

Score	Meaning	Example
1	Very minor	Small defect; function is almost normal.
2	Minor	Limited malfunction; most of the asset still works.
3	Moderate	Noticeable malfunction affecting normal use.
4	Serious	Major failure; use or service is substantially affected.
5	Critical	Complete failure or severe damage.

Urgency

Score	Meaning	Typical interpretation
1	Very low	Can wait; no immediate operational impact.
2	Low	Can be handled during routine maintenance.
3	Moderate	Needs planned attention in a reasonable time.
4	High	Needs prompt attention, often the same day.
5	Immediate	Needs immediate response or escalation.

Safety Risk

Score	Meaning	Typical interpretation
0	None	No identified safety risk.
1	Very low	Negligible or minimal risk.
2	Low	Limited risk that still needs attention.
3	Moderate	Meaningful risk requiring precautions.
4	High	Serious risk; prompt control is needed.
5	Critical	Immediate danger to people or major building safety.

Affected People (AP)

Score	Peoples
1	Below 25
2	25< AP <100
3	100 < AP < 200
4	200 < AP <1500
5	Above 1500

Location Importance (LI)

Score	Zone	Location Importance
1.5		Garden
2	A	Quarters
2.5	A	Guard Room
3	A	TAG
3.5	C	Canteen 1 and 2
4	C	Indoor
4.5	C	Gym
5	A	Civil Canteen
5.5	C	Medical Center
6	A	Admin Building
6.5	A	Shopping Complex
7	B	Civil Engineering Division
7.5	B	Textile and Clothing Technology Division
8	B	Maritime Engineering and Nautical Science Division
8.5	B	Mechanical Engineering Division
9	B	Chemical Engineering and Polymer Technology Division
9.5	B	Electrical and Electronic Engineering Division
10	C	Hostel

Waiting Time (WT)

Time Period	**Score**
First 24 hours	0
24 hours to 72 hours	1
72 hours to 96 hours	2
96 hours to 120 hours	3
120 hours to 168 hours	4
After week	5

5. Priority Levels

Priority level	Score range	General interpretation
CRITICAL	76–95	Immediate response; emergency override can also force this level.
HIGH	56–75	Prompt action, often same-day depending on local policy.
MEDIUM	31–55	Schedule and monitor after higher-priority requests.
LOW	0–30	Routine maintenance when resources are available.